ARE WE DOOMED TO DROWN IN OUR OWN PLASTIC WASTE, DESPITE DECADES OF RECYCLING EFFORTS?

Once celebrated for its durability and versatility, plastic has become more of a curse than a blessing. Integrated into every part of modern life, it's now polluting our oceans, poisoning wildlife, and threatening human health. Our attempts at managing this synthetic material have been futile. Yet science-unpredictable and boundless-has finally delivered a breakthrough in the crisis we've long struggled to control.

Most conventional plastics are made from petroleum, a fossil fuel formed from the remains of ancient organisms over millions of years. While its origins are biological, the industrial process transforms it into a substance that natural decomposers don't recognize or easily break down. As a result, they are non-biodegradable, which makes them durable. Although plastics can slowly break down over extremely long, this isn't true biodegradation like what happens with organic materials. In recent years, the growing amount of plastic waste has become a significant environmental concern. Their prolonged presence in the environment—especially in marine ecosystems— has disrupted the balance of nature. More recently, microplastics have been identified as a potential threat to human health.

One major issue is that when plastic is recycled, it loses quality. To create new products, fresh plastic must often be added to the recycled material. More importantly, plastic cannot be recycled indefinitely. After a few cycles, it becomes unrecyclable. Today, less than 10% of the plastic we produce is actually recycled, making recycling appear more like a symbolic gesture than a real solution.

Yet plastics are deeply integrated into modern life—they're used in everything from clothing and smartphones to food packaging and medical equipment. We cannot simply stop manufacturing plastic products. So, is there no solution to this plastic curse? — there seems a ray of hope.

Just when it seemed like we are doomed in this situation nature showed up once and again and astonished us in unexpected ways. In 2016, a team of Japanese researchers from Kyoto Institute of Technology and Keio University made a revolutionary discovery from samples from plastic recycling plant in Sakai City. They stumbled upon something extraordinary – a bacterium called Ideonella sakaiensis. These tiny marvels of nature eat plastic and harvests carbon in plastic for energy which they use to grow move and reproduce. Plastics are made by linking monomers into long polymer chains. Ideonella sakaiensis produces two enzymes, PETase and MHETase—that function like molecular scissors. These enzymes help break down these plastic polymers into their monomers. The fact that I. sakaiensis survives solely on PET suggests that it evolved specifically to degrade this synthetic material which show how these microorganisms have evolved to survive in this environmental condition.

Then, in April 2020, another breakthrough emerged. A research team from the University of Toulouse and the French biotech company Carbios announced a mutant enzyme that could break down 90% of a PET plastic bottle in just 10 hours. Because this process doesn't just destroy plastic—it deconstructs it.

The plastic bottle is broken down into its fundamental chemical building blocks. These monomers can then be reused to create new plastic, theoretically enabling true recycling with no quality loss. While the process does involve some mechanical pre-treatment to chop the plastic into smaller bits, the enzymes still deserve most of the credit.

These discoveries reveal a broader truth: nature's biological ingenuity knows no limits. From PET (Polyethylene Terephthalate) bottles to polythene bags, enzymes are now breaking down plastics like never before.

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